The slope formula

1 Determine the slope of the line that passes through each of the following pairs of points, using $m = \frac{\text{rise}}{\text{run}}$ or $m = \frac{y_2 - y_1}{x_2 - x_1}$.

a Point $A\,(-1, 0)$ and Point $B\,(0, 3)$

b Poit $A\,(2, -6)$ and the origin

c Point $A\,(4, 1)$ and Point $B\,(11, -6)$

d Point $A\,(-3, -1)$ and Point $B\,(-5, 1)$

e Point $A\,(0, -4)$ and Point $B\,(5, -1)$

f Point $A\,(-7, -6)$ and Point $B\,(-5, -8)$

g Point $A\,(-3, 4)$ and Point $B\,(1, 4)$

h Point $A\,(3, 4)$ and Point $B\,(5, 4)$

2 If we have two points and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)? Explain.

3 Determine whether the each of the following pairs of coordinates will have a slope that is defined or undefined.

a $(-10, 5)$ and $(-10, 12)$

b $(10, 5)$ and $(10, 1)$

c $(10, -1)$ and $(-10, -1)$

d $(-10, 5)$ and $(10, 5)$

e $(10, 7)$ and $(10, 2)$

f $(10, -2)$ and $(-10, -2)$

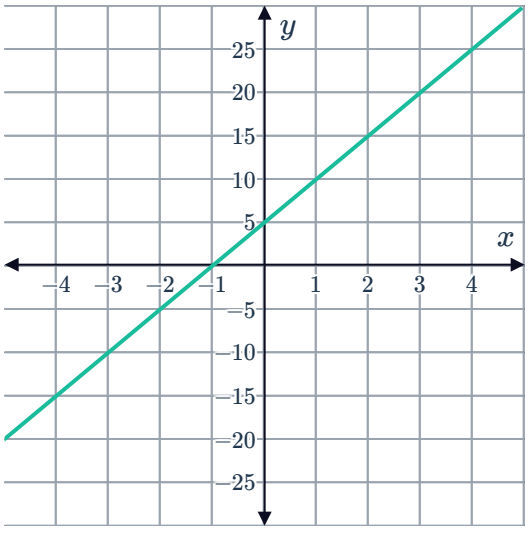
g $(-10, 7)$ and $(-10, 12)$

h $(-10, 7)$ and $(10, 7)$

4 Consider the line $y = 5x + 5$ that has been graphed on the number plane.

a Find the y value of the point on the line where $x = 4$.

b Using $m = \frac{\text{rise}}{\text{run}}$, the point found in the previous part and the y-intercept, find the slope of the line.



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5 Given the slope of the line passing through the two points, find the value of the variable:

a $(4, -3)$ and $(1, t)$, slope $= -2$

b $(5, 3)$ and $(2, t)$, slope $= -4$

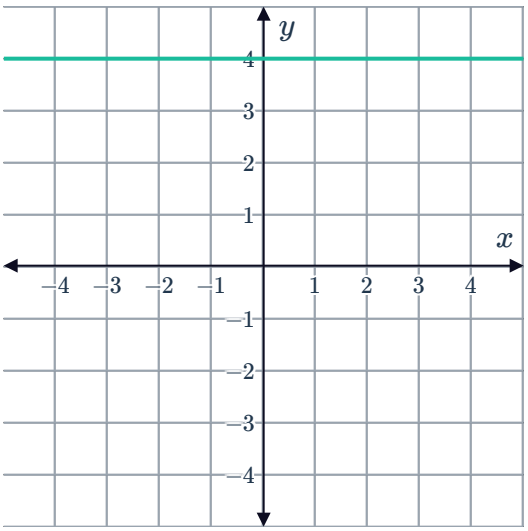
c $(5, 3)$ and $(d, 63)$, slope $= 4$

d $(11, c)$ and $(-20, 16)$, slope $= -\frac{4}{7}$

e $(1, 2)$ and $(-4, t)$, slope$= 5$

6 A paratrooper falls to the ground along a diagonal line. His fall begins 1157 m above the ground, and the line he follows has a slope of 1.3. That is, he falls 1.3 m vertically for every 1 m he moves across horizontally. How far horizontally across the ground does he land from his initial position in the sky?

7 Consider the following graph:

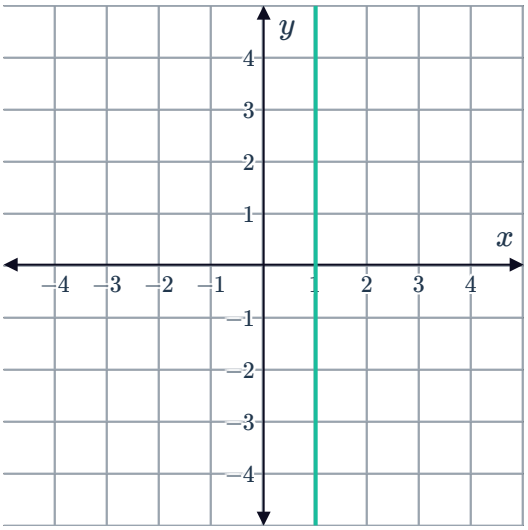


a Determine the slope of the line.

b Find the y-value of the y-intercept of the line.

c Does this line have an x-intercept?

8 Consider the following graph:



a Determine the slope of the line.

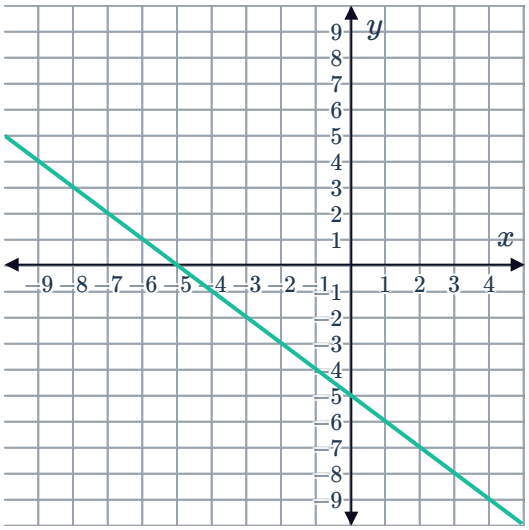
b Does this line have a y-intercept?

c Identify the x-intercept of the line.

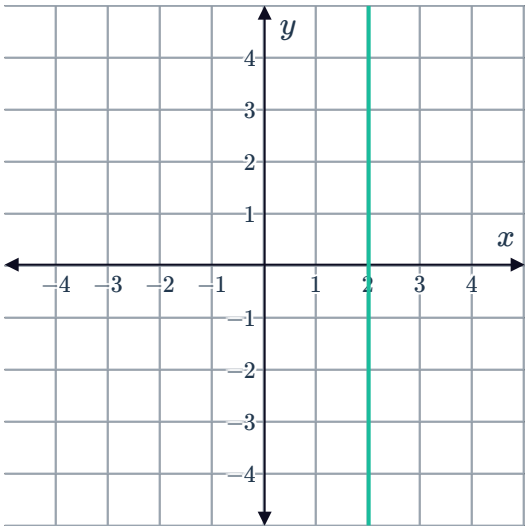
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9 Consider the following graphs.
Determine whether the the slope is 0, undefined or neither.

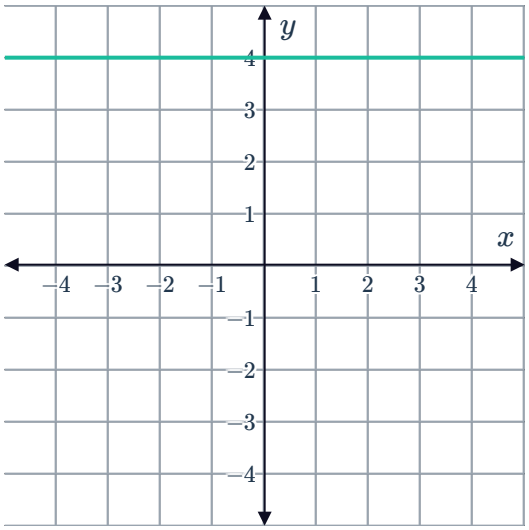
a



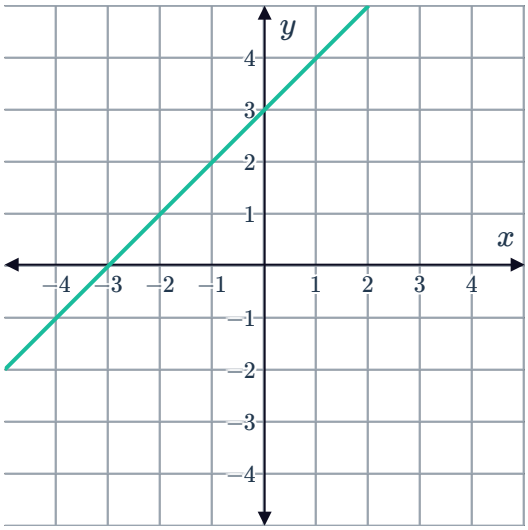
b



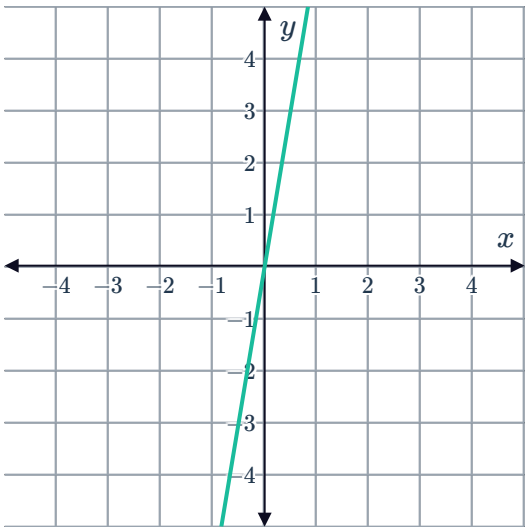
c



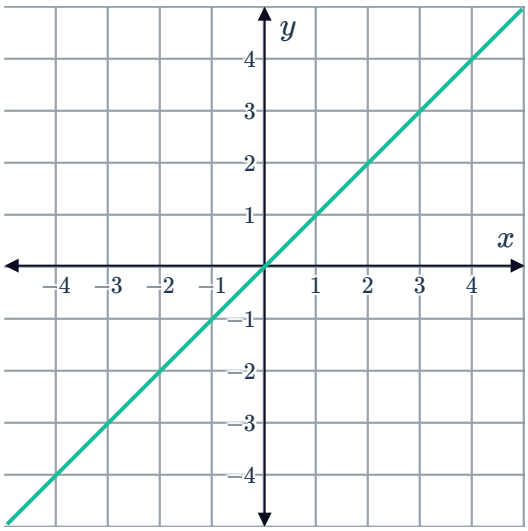
d



e

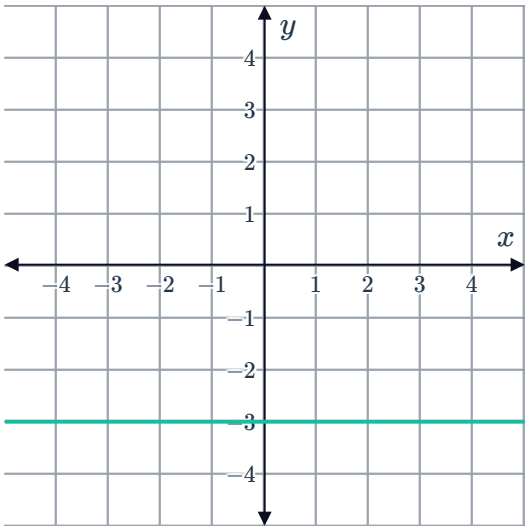


f



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g



10 Explain what it means if the slope of a line is:

a undefined

b zero

11 State whether each of the following statements is correct for vertical lines.

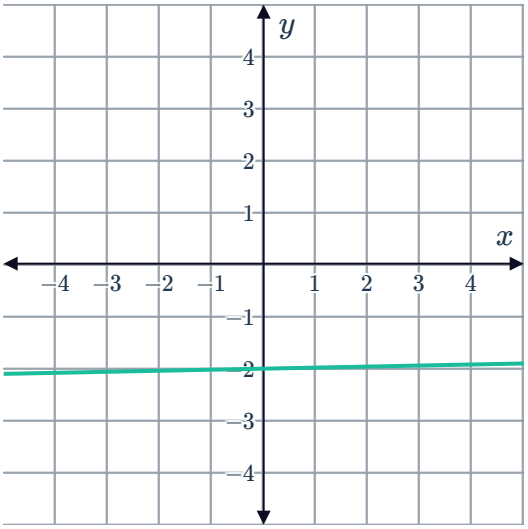
a A vertical line's slope is very large because it is so steep.

b A vertical line has a slope that is undefined because, when calculating the slope using the formula $\frac{\text{rise}}{\text{run}}$, the run is 0 and it is not possible to divide a number by 0.

c A vertical line's slope is equal to 0 because when calculating the slope using the formula $\frac{\text{rise}}{\text{run}}$, the run is 0.

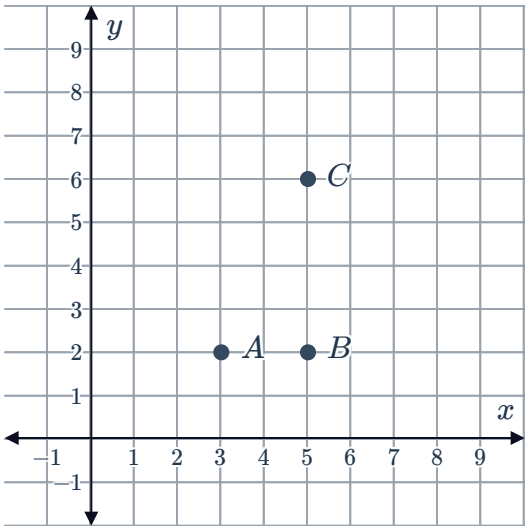
12 The line in the graph is nearly horizontal. Which of the following is most likely to be its slope?

A 0.33 **B** 0.03 **C** 3 **D** 0.3



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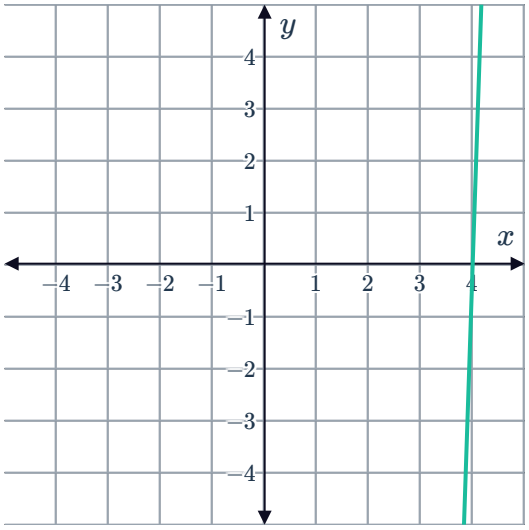
13 Consider the points on the graph.



a Determine the slope of $\overline{AB}$.

b Determine the slope of $\overline{BC}$.

14 The line that has been graphed is nearly vertical.
Which of the following is most likely to be its slope?



A 5 **B** 55 **C** undefined **D** 0.55

15 The function $f(t) = -25t^2 + 30t + 5$ gives the height of a ball, in feet, t seconds after it is thrown.
Determine the average rate of change, in feet per second, over the interval $(0.75, 1.25)$.

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Additional questions

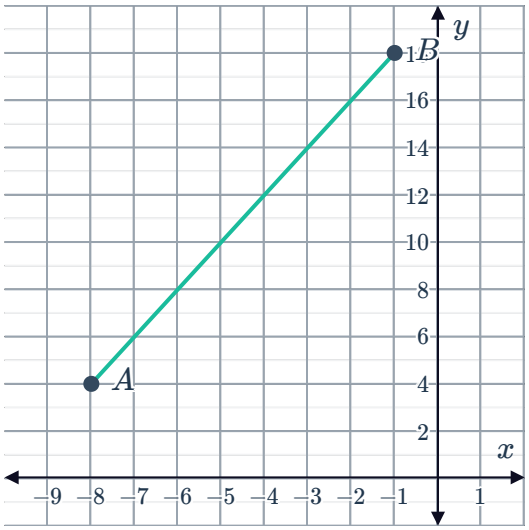
16 For each of the following intervals between points A and B:

i Find the rise.

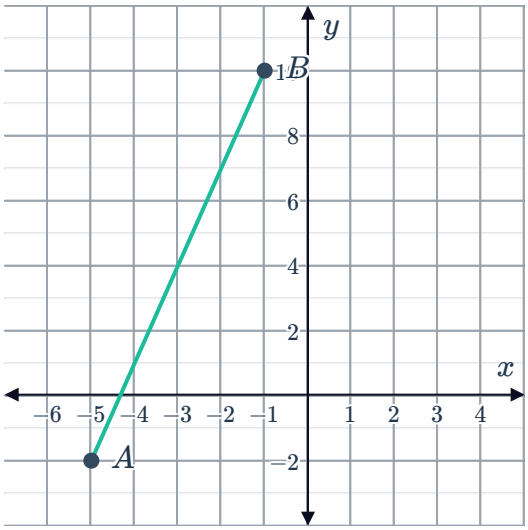
ii Find the run.

iii Find the slope.

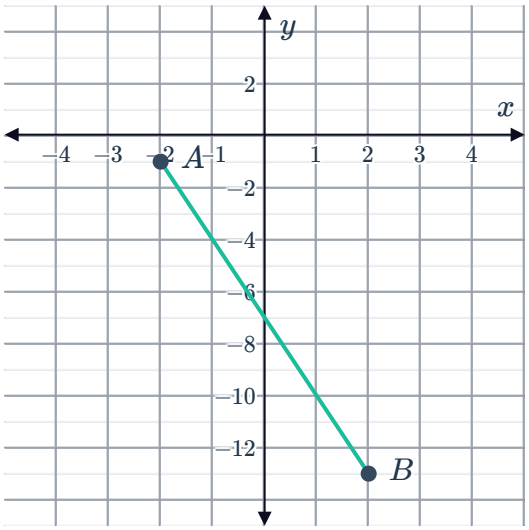
a $A\,(-8, 4)$ and $B\,(-1, 18)$



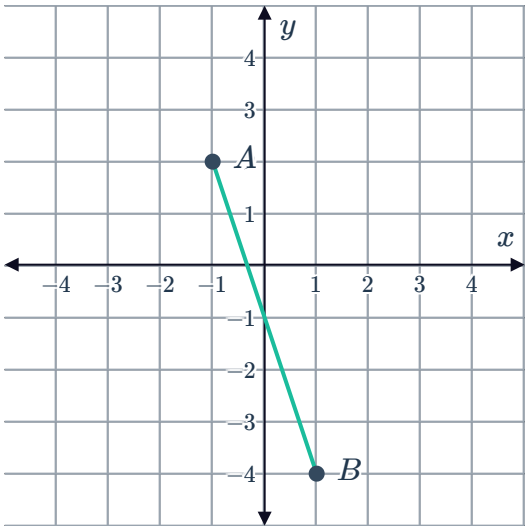
b $A\,(-5, -2)$ and $B\,(-1, 10)$



c $A\,(-2, -1)$ and $B\,(2, -13)$



d $A\,(-1, 2)$ and $B\,(1, -4)$



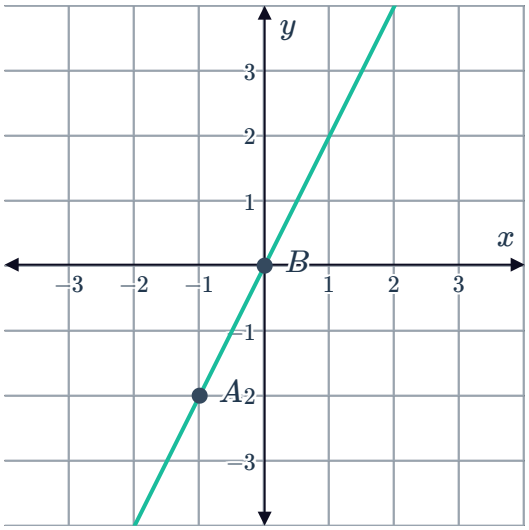
17 Explain why a vertical line has an undefined slope.

18 State the slope of any line parallel to the x-axis.

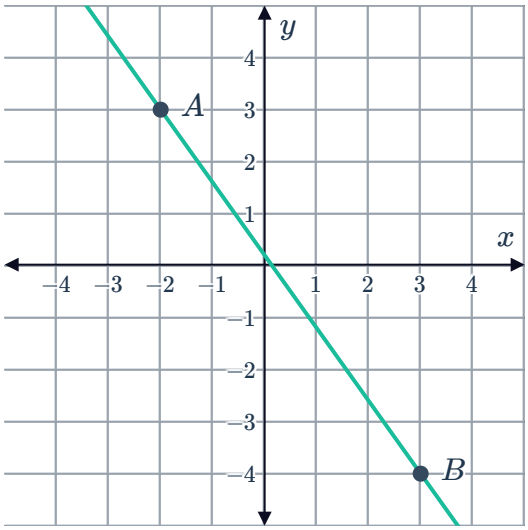
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19 Determine the slope of the following lines passing through points A and B:

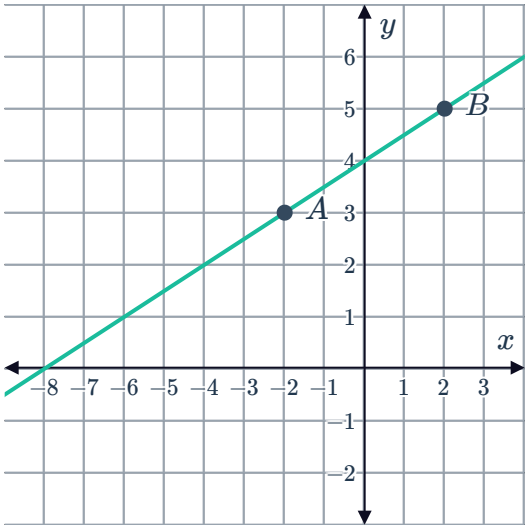
a



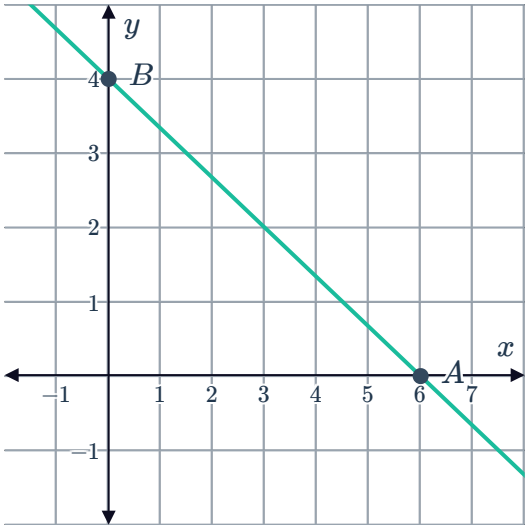
b



c



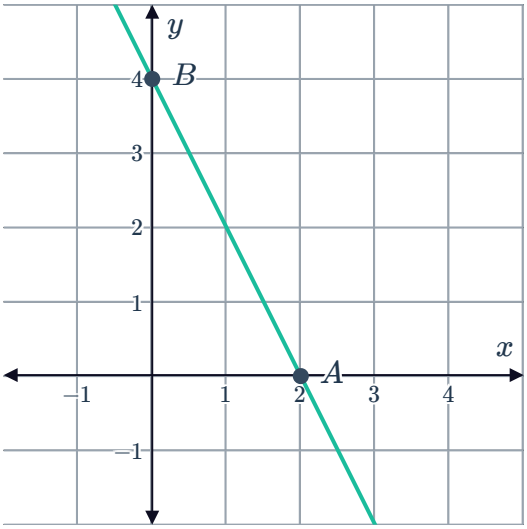
d



20 Consider the line plotted, where $A\,(2, 0)$ and $B\,(0, 4)$ both lie on the line.

a Determine the slope of the line.

b As x increases, what happens to the value of y?



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21 Determine the slope of the line that passes through the given points:

a $(-1, 0)$ and $(0, 3)$

b $(-4, 7)$ and $(1, 10)$

c $(1, -4)$ and the origin

d $(2, -6)$ and the origin

e $(6, 4)$ and $(3, 4)$

f $(-6, 5)$ and $(4, 5)$

g $(-2, -5)$ and $(-9, -12)$

h $(-3, -1)$ and $(-5, 1)$

22 Consider the points $A(-11, -9)$, $B(-5, 1)$ and $C(-2, 6)$:

a Determine the slope of AB.

b Determine the slope of BC.

c Do the points A, B and C lie in a straight line?

23 Consider the following points: $A(26, m - 24)$, $B(-1, m)$ and $C(-10, 9)$. Find m, given that A, B and C are collinear.

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